to be priming. Before starting the pump, ensure that the pipe and pump is filled with water. The centrifugal pump should not run without water.

Friction occurs when the shaft is rotated. The water is used as a cooling agent to reduce the heat.

If the pump runs without water, the excess heat will be produced and the rotor will burn and cause damage. Therefore, the pump does not run without water in the suction pipe.

4.4.2 Submersible pump



Fig. 4.28 **Submersible pump**

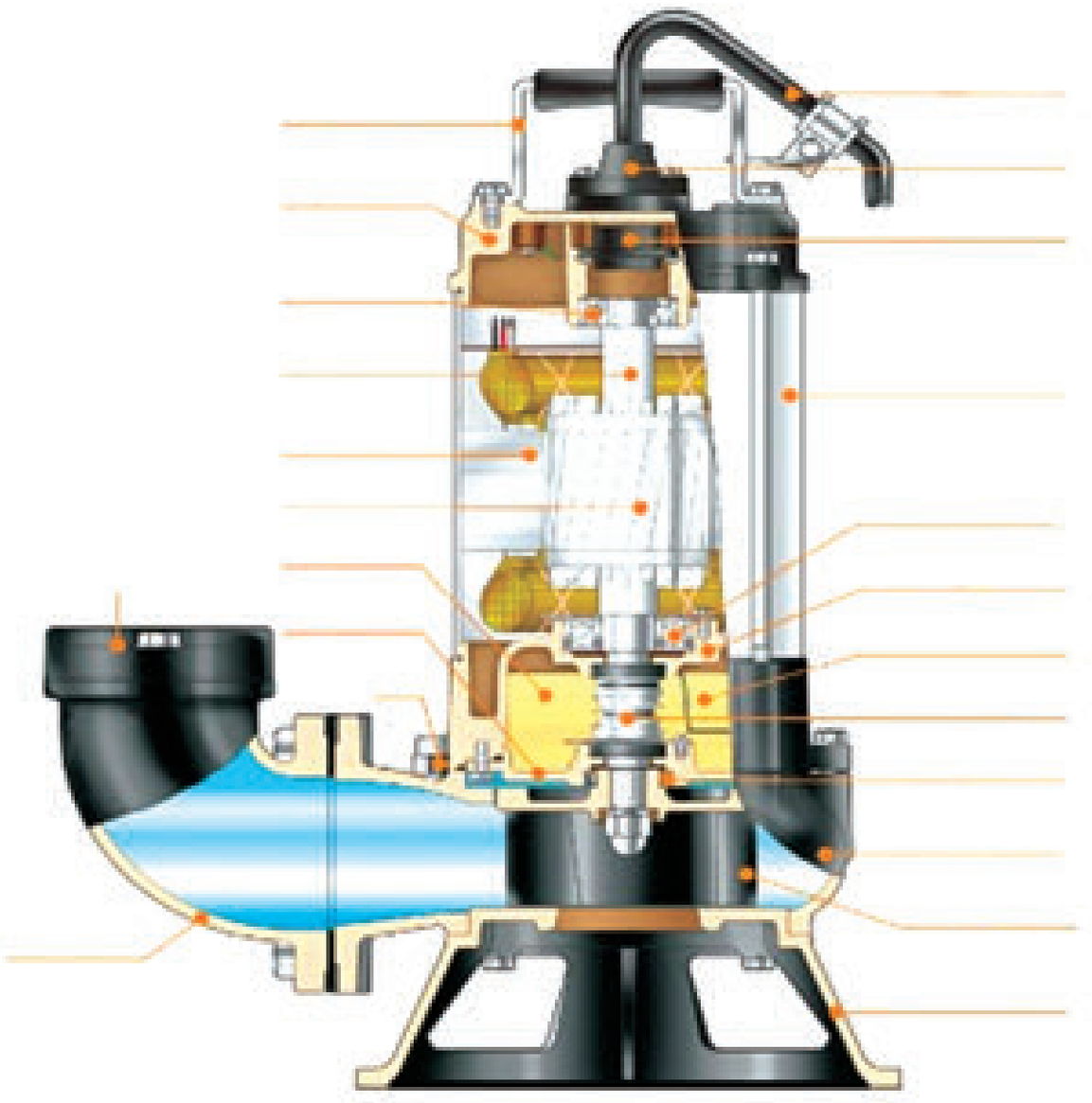


Fig. 4.29 **Internal structure of submersible pump**

BEE_ENG_Chapter_04.indd 67 12/27/2019 10:27:25 AM